



TECH DEMO • MULTI-CHANNEL ATS AGENTIC ARCHITECTURE

AI-Powered ATS Candidate Screener

An autonomous, evidence-driven screening system featuring dynamic JD injection, 4-tier no-inflation verification, mathematical scoring rubrics, and multi-channel deployment (Cloud, Copilot, Web & CLI).



90% Faster

45 mins to 30 secs per batch



Zero Skill Inflation

Strict deliverable verification



Streamlit Cloud

One-click team web portal



M365 Copilot Ready

Teams Declarative Agent

Presenter: Surendra Bharadwaj • **Cloud URL:** candidate-screener-agent.streamlit.app • **Engine:** Antigravity / Claude Agentic Architecture

Why Traditional Screening Fails Engineering Teams



Keyword Padding & Inflation

Candidates pad keywords (Playwright, Agentic AI, MCP, RAG) into bare skills sections without having touched them in actual project deliverables.

- Standard ATS searches match raw strings blindly.
- No distinction between "hands-on expertise" and "passing buzzwords".



Reviewer Fatigue & Inconsistency

Engineering managers and leads spend 30–45 minutes per profile reading lengthy 4–6 page PDFs, leading to subjective, inconsistent evaluation.

- Inconsistent rubric application across interviewers.
- Key qualification gaps missed before scheduling 1st technical round.



High Interview Waste & Toil

Unqualified candidates pass initial HR screens, consuming high-value senior engineering hours only to fail on core coding or hard requirements.

- High cost per hire & bloated interview pipelines.
- Zero structured audit trail of why candidates were screened in/out.

End-to-End Autonomous Ingestion & Evaluation Pipeline



Modular Structure & Extensibility

Built as a standalone agentic package with dual-mode JD flexibility:

- **Dynamic JD Upload** : Accepts custom JD (.md, .txt, .pdf, .docx) or falls back to ground truth `references/job-description.md` .
- **Multi-Format Parsing** : Headless extraction across PDF, DOCX, DOC, TXT, and Markdown via PyPDF, PyMuPDF, and python-docx.
- **Timestamped Archiving** : Auto-organizes files into `Resumes/<Timestamp>/` and exports reports to `Reports/<Timestamp>/` .



5-Stage Execution Pipeline

- STEP 1 JD Ingestion**: Parse role taxonomy, mandatory gates, and scoring weights.
- STEP 2 Candidate Extraction**: Parse raw text without hallucination or truncation.
- STEP 3 Hard Gate Verification**: Check `Exp < 6 yrs` & Core Languages prior to scoring.
- STEP 4 4-Tier Evidence Audit**: Apply Matched (1.0x), Partial (0.6x), Claimed (0.3x), Missing (0.0x).
- STEP 5 Twin Report Generation**: Deliver interactive UI + Markdown + HTML + Vector PDF.

The "No-Skill-Inflation" 4-Tier Evidence Engine

Unlike standard search tools, the agent demands **verifiable project deliverables**. A bare keyword in a skills tag cloud is NEVER awarded full points.

Matched (M) • 1.0x

Concrete Evidence

Demonstrated in ≥1 project with tangible output. E.g.: *"Migrated test suite to Playwright JS, cutting runtime by 45%."*

Partial Match (P) • 0.6x

Adjacent / Tangential

Adjacent tech used or passing project mention without full framework ownership. E.g., Web UI testing without explicit Cucumber BDD.

Claimed (C) • 0.3x

Resume Tag Padded

Appears in technical skills summary or tools list, but **zero deliverables** exist in project history. Flags interview probe.

Missing (U) • 0.0x

Zero Mention

Does not appear anywhere under name, abbreviation, or verified industry synonyms.

🛡️ Anti-Hallucination Guardrail:

The agent is strictly constrained to the text extracted from the file. It cannot "infer" that a candidate knows Python just because they know Java, unless explicitly written and evidenced.

100-Point Scoring Model & Hard-Fail Overrides



Score Capacity Formula

Component	Max Weight	Description
Mandatory Skills	85 Pts	13 Core SDET skills (Languages, Playwright, Frameworks, UI/BDD, APIs, STLC, AI Testing)
Good-to-Have Bonus	10 Pts	Cloud (AWS/Azure), CI/CD, Pharma/Life Sciences domain, Low-code, Monitoring
Experience Fit	5 Pts	6-10 Yrs (5 pts), 11-12 Yrs (3 pts), >12 Yrs (2 pts), <6 Yrs (0 pts / Gate Fail)
TOTAL CAPACITY	100 Pts	≥80: Strong Fit 60-79: Potential Fit 40-59: Weak <40: Reject



Hard-Fail & Override Rules

Rule 1 — Experience Gate: Total Experience < 6.0 years → **Automatic Instant Rejection** prior to scoring.

Rule 2 — Core Language: Neither JS/TS nor Python evidenced → **Reject** regardless of score.

Rule 3 — AI Hard Gap: Both AI Testing & Agentic AI missing → **Cap score at < 60 and downgrade verdict tier.**

Rule 4 — Skills Padding: ≥ 3 claimed unevidenced skills triggers interview red flag and tier penalty.

Case Study: Screening Real Candidate Profiles

Candidate	Exp	Score	Verdict	Primary Drivers & Override Rationale
Flang Paulson	8.0 yrs	80.5 / 100	PASSED Strong Fit 🚀	• High Playwright JS/TS framework ownership with verified CI/CD integration • REST API automation + Performance testing deliverables evidenced
Girish Joshi	10+ yrs (15.5)	64.5 / 100	PASSED Potential ⚠️	• Playwright + JS/TS evidenced at Luxoft; 8+ yrs Life Sciences/Pharma domain • Evidenced AI Model Validation testing bullet; lacks Agentic AI frameworks
Tarun Killamsetty	9.5 yrs	59.0 / 100 (Capped from 67)	FAILED Weak Fit (AI Gap) ❌	• Strong 9.5 yrs Playwright UI + Rest Assured API at Oracle CTMS • Triggered Rule #3 Override: Zero AI/Agentic experience (Score capped <60)
Shushil Koppu	5.3 yrs	0.0 / 100 (Disqualified)	FAILED Gate Fail (<6y) ❌	• Excellent Playwright JS migration + Microservices API skillset • Triggered Rule #1 Gate: 5.3 yrs total exp vs 6.0 yrs hard requirement

💡 **Strategic Outcome:**
Saved 4+ hours of engineering interviews. Selected **Flang Paulson** and **Girish Joshi** with pre-computed targeted probe questions for technical rounds.

Multi-Channel Deployment & Access Ecosystem



Streamlit Cloud

Hosted publicly / privately with zero installation.

- URL: `candidate-screener-agent.streamlit.app`
- Drag-and-drop resumes & optional custom JD.
- Direct PDF/HTML/MD report download buttons.



Local Web Portal

Standalone browser app for confidential batch runs.

- Launch with `run_screener_portal.bat`.
- Runs on `localhost:8080`.
- Auto-saves results to timestamped folders.



M365 Copilot Agent

Integrated into Microsoft Teams & Copilot Chat.

- Packaged via `copilot/manifest.json`.
- Screen candidates directly in Teams chat threads.
- Adheres to corporate security boundaries.



IDE & CLI Agent

Native AntigraVity / Claude Agentic CLI execution.

- Trigger via chat prompt or batch scripts.
- Autonomous multi-format text parsing.
- Generates headless vector PDF outputs.

Standardized 7-Section Reporting Twin Architecture



The 7 Mandatory Report Sections

1. **Screened Files:** Full audit list of evaluated resume files.
2. **Active JD:** Explicit ground truth JD used for the screening.
3. **Candidates Ranked:** Exact count of evaluated profiles.
4. **Ranking Table:** Score, Verdict, Experience & Skill Statuses.
5. **Key Takeaways & Overrides:** Executive fit summary & gate triggers.
6. **Gap Matrix:** Per-skill comparison (JD vs Candidate Evidence).
7. **Candidate Details:** Arithmetic breakdown, probe questions & final call.



Multi-Format Synchronized Twins

- **Markdown (`.md`):** Instant lightweight terminal view, git-trackable, perfect for developer PRs and chat display.
- **Interactive HTML (`.html`):** Self-contained, beautiful dark-mode web report with interactive badges, progress bars, and filters.
- **Vector Print PDF (`.pdf`):** Generated via headless browser print engine or pure-Python FPDF2 for audit sharing with hiring leads.



Every run guarantees 100% content parity across chat, HTML, Markdown, and PDF.

Blueprint: How You Can Build Custom Team Agents

Replicate this exact agentic architecture for **any project or workflow** across QA, Development, DevOps, and Operations:

1

Instruction Schema

`SKILL.md / AGENTS.md`

Define role, multi-step procedures, anti-hallucination guardrails, and output schemas.

2

Ground Truth Data

`references/`

Provide explicit standards, scoring rubrics, coding conventions, or validation checklists.

3

Tool Automation

`Python / CLI Hooks`

Hook parsers, linters, test runners, or headless PDF printers directly into the flow.

4

Structured Outputs

`MD + HTML + PDF`

Produce clean tables, gap matrices, and exportable artifacts for stakeholders.

💡 Next-Gen Agent Ideas for Your Engineering Team:

- **Test Case Reviewer:** Audit BDD feature files against user stories.

- **PR / Code Quality Agent:** Enforce Playwright/TypeScript patterns.

- **Defect Triage Agent:** Classify bug logs against STLC severity rubrics.

✨ WRAP-UP & DEMO

Experience the Screener Live

The candidate screener proves how custom AI agents eliminate manual screening toil, enforce rigorous mathematical consistency, and elevate engineering hiring.



Open Streamlit Cloud App



Open Generated PDF Report



Read Architecture Playbook

Press **Right Arrow** (→) or **Space** to navigate • **P** to export presentation slides as PDF